

1 Solution — GIAM 3.1

Problem 8

1.1 Problem

Prove that for all integers a and b , if a is odd and $6 \mid (a + b)$, then b is odd.

1.2 Proof

***Proof.** Let a and b be integers, and suppose a is odd and $6 \mid (a + b)$. By the definition of odd there is an integer m with $a = 2m + 1$, and by the definition of divides there is an integer n with $a + b = 6n$. Solving for b ,*

$$b = 6n - a = 6n - (2m + 1) = 6n - 2m - 1 = 2(3n - m - 1) + 1.$$

Now $3n - m - 1$ is an integer, since $\mathbb{Z}$ is closed under multiplication and subtraction. So $b = 2k + 1$ with $k = 3n - m - 1 \in \mathbb{Z}$, and hence b is odd. ■

Two hypotheses, two definitions unpacked, then solve for b and regroup so a factor of 2 sits out front with a lone $+1$ left over.

1.3 Reading the Proof Structurally

Two hypotheses, both unpacked into equations; then solve for b and group to expose a factor of 2.

The direct proof

$$a \text{ is odd} \Rightarrow a = 2m + 1, m \in \mathbb{Z}$$

$$6 \mid (a + b) \Rightarrow a + b = 6n, n \in \mathbb{Z}$$

$$b = 6n - a = 6n - (2m + 1) = 2(3n - m - 1) + 1$$

$$3n - m - 1 \in \mathbb{Z}, \text{ so } b = 2k + 1 \text{ with } k = 3n - m - 1$$

hence b is odd ■

The same proof, read structurally

$$2 \mid 6 \text{ and } 6 \mid (a + b) \Rightarrow 2 \mid (a + b)$$

transitivity of divisibility

so $a + b$ is even, and a is odd

from the parity table: odd + even = odd, so for an even sum b must be odd

Which divisors would work in place of 6?

divisor d	is d even?	does the theorem hold?
2	yes	holds
3	no	fails — $a = 1, b = 2$
4	yes	holds
5	no	fails
6	yes	holds — this problem
7	no	fails
8	yes	holds

Only the factor of 2 in the divisor does any work — the 3 in 6 is never used.

Figure 1.1: Top panel, the direct proof: a is odd gives a equal to $2m$ plus 1 with m an integer; 6 divides a plus b gives a plus b equal to $6n$ with n an integer. Then b equals $6n$ minus a equals $6n$ minus $(2m$ plus 1) equals 2 times $(3n$ minus m minus 1) plus 1. Since $3n$ minus m minus 1 is an integer, b equals $2k$ plus 1, hence b is odd. Second panel, the same proof read structurally: 2 divides 6 and 6 divides a plus b , so 2 divides a plus b by transitivity of divisibility; hence a plus b is even, and a is odd; and from the parity table, odd plus even is odd, so for an even sum b must be odd. Below, a table asking which divisors would work in place of 6. Divisor 2, even, holds. Divisor 3, not even, fails with a equals 1 and b equals 2. Divisor 4, even, holds. Divisor 5, not even, fails. Divisor 6, even, holds — this problem, highlighted. Divisor 7, not even, fails. Divisor 8, even, holds. Only the factor of 2 in the divisor does any work; the 3 in 6 is never used.

The algebra conceals a simpler story. Since $2 \mid 6$ and $6 \mid (a + b)$, transitivity of divisibility gives $2 \mid (a + b)$ — so $a + b$ is **even**. Now consult the parity table of Problem 4: with a odd, the sum $a + b$ is even only when b is odd. Done.

Both routes are correct; the second explains *why* the theorem is true rather than merely verifying it.

1.4 Remark — The Hypothesis Is Much Stronger Than Needed

Notice what the proof never used: **the factor of 3 in 6**. Only $2 \mid 6$ mattered. So the theorem remains true with the far weaker hypothesis $2 \mid (a + b)$, and equally with 4, 8, 12, or any even divisor:

divisor d	d even?	does the theorem hold?
2	yes	holds
3	no	fails — $a = 1, b = 2$
4	yes	holds
5	no	fails
6	yes	holds (this problem)
7	no	fails
8	yes	holds

Table 1.1

The pattern, confirmed by exhaustive check on $[-60, 60]^2$, is that the theorem holds **exactly when the divisor is even**. For an odd divisor the hypothesis says nothing about parity: $a = 1$ is odd and $3 \mid (1 + 2)$, yet $b = 2$ is even.

1.5 Remark — Why Stating a Stronger Hypothesis Is Legitimate

It might seem a defect that the problem assumes $6 \mid (a + b)$ when $2 \mid (a + b)$ would do. It is not an error — a theorem with a stronger hypothesis is simply a **weaker theorem**, and weaker theorems are still true.

There are ordinary reasons to state one. The stronger hypothesis may be what is actually available in the situation you care about; it may make the statement read more naturally; or, as here, it may be a deliberate exercise in noticing that not every assumption gets used. What *would* be a defect is claiming the hypothesis is necessary when it is not.

The diagnostic is the one from Problem 5: **point to the line where each hypothesis is used**. Here " a is odd" is used to produce $a = 2m + 1$, and " $6 \mid (a + b)$ " is used to produce $a + b = 6n$ — but the 6 could have been any even number and the algebra would still close.

1.6 Remark — Unpacking Two Definitions at Once

This is the first problem in the section with **two** hypotheses to unpack, and each contributes its own existential witness:

$$\underbrace{a \text{ odd}}_{\Rightarrow \exists m, a=2m+1} \quad \text{and} \quad \underbrace{6 \mid (a + b)}_{\Rightarrow \exists n, a+b=6n}.$$

As always the two witnesses need **different letters** — writing n for both would silently assume a relationship that was never given. And notice the definition of *divides* is itself existential: $d \mid x$ means $\exists q \in \mathbb{Z}, x = dq$. Recognising that “divides” unpacks the same way “odd” and “even” do is the small conceptual step this problem adds.

1.7 Remark — Solving for the Right Variable

The computation solves for b rather than manipulating $a + b$ directly, and that choice is what makes the proof short. The conclusion is a statement *about* b , so the goal is an expression for b alone in the shape $2(\text{integer}) + 1$.

This is worth stating as a heuristic: **look at what the conclusion is about, and isolate that quantity**. A proof that instead expanded $a + b = 2m + 1 + b = 6n$ and tried to reason about the whole sum would work eventually, but only after rearranging to $b = \dots$ anyway. Aiming the algebra at the target from the start saves the detour.

1.8 Remark — The Contrapositive Route

An alternative proof runs by contraposition (Problem 2.2.8). The contrapositive is:

If b is even, then a is even or $6 \nmid (a + b)$.

Suppose b is even and a is odd. Then $a + b$ is odd (Problem 4), so $a + b$ is not divisible by 2 , hence not by 6 . That establishes the contrapositive and therefore the theorem.

This version is arguably more transparent, since it shows the parity clash directly. But note it requires negating a conjunction into a disjunction and then handling the cases — the direct proof avoids that bookkeeping entirely. When both routes are available, the direct one is usually shorter for a conclusion this simple.